Brain Interacts
A PEOPLE ANALYTICS COMPANY

WHITEPAPER

The ROI of Psychological Safety

A behavioral approach to high-velocity innovation.

The Behavioral Enterprise Series

Brain Interacts

Written for

CTOs · Chief Product Officers · Heads of R&D and Innovation · Engineering and Product Leaders

AT A GLANCE

Four ideas in one page

The problem

Bad news travels slowly. Risks that a few people see in month two become launch delays and rework in month nine.

The insight

Silence is a rational response to interpersonal risk, not a character flaw. Teams speak up when the social cost of doing so is low, and leaders control that cost.

The method

The Safety Engine: a five-step loop (Frame, Invite, Respond, Embed, Measure) that builds psychological safety through repeatable leader behaviors and team routines.

The payoff

Earlier detection of problems, less rework, faster learning cycles and better retention, which can be modeled in financial terms.

THE MEASUREMENT TRAP

A low count of reported problems can mean a safe operation or a silent one.

THE SCENE

The problem that was visible in month two

A flagship product misses its launch date by a quarter. The post-mortem is uncomfortable for a specific reason: the root cause was not a surprise.

MONTH TWO
A senior engineer had doubts about the architecture

MONTH FOUR
A tester saw inconsistent results

PILOT
A product manager sensed the customer pilot was lukewarm

LAUNCH
The status report stayed green until it turned red

Each person had a piece of the picture. None raised it in a forum where it could change the plan.

This pattern is common enough to have a name: the "watermelon project," green on the outside and red on the inside. It is expensive because the cost of fixing a problem rises the longer it stays hidden.

> *The cause is rarely incompetence. It is the cost of speaking up.*

THE EVIDENCE

Silence is common, and it hides in the data

85%

Milliken, Morrison & Hewlin, 2003

In an exploratory interview study of 40 employees, 85% had been in situations where they felt unable to raise an important issue with a supervisor. The most common reason was fear of being viewed negatively and damaging relationships.

The sample is small and the study qualitative, so treat 85% as a signal that silence is common rather than as a population rate.

Even so, it should unsettle any leader who equates a quiet meeting with a healthy project.

An older finding sharpens the point. In her study of hospital units, Edmondson (1996) found that the units with the best leadership and coordination reported *more* medication errors, not fewer. The better units were not making more mistakes; they were more willing to report them.

THE MEASUREMENT TRAP

A low count of reported problems can mean a safe operation or a silent one. Without psychological safety, the data a leader relies on is the data people were willing to share.

For organizations that depend on R&D and product development, this matters directly. Innovation is a process of making uncertain bets, discovering that some are wrong and adjusting quickly. The speed of that cycle depends on how quickly people surface bad news, challenge assumptions and admit what they do not know. Psychological safety is the condition that makes those behaviors normal.

THE SCIENCE

Why smart people stay quiet

DEFINITION

Psychological safety is a shared belief held by members of a team that the team is safe for interpersonal risk-taking (Edmondson, 1999).

It is about the *team climate*, not about individual personality, and it is not the same as comfort or trust between friends. It describes whether people expect to be punished or embarrassed for asking a question, admitting a mistake, offering an idea or disagreeing.

Why people stay silent

Every act of voice carries a small social risk:

Looking ignorant	**Looking disloyal**	**Damaging a relationship**	**Being labeled difficult**

Detert and Edmondson (2011) describe "implicit voice theories": taken-for-granted beliefs, formed early in careers, such as "it is risky to challenge a superior" or "you should not embarrass the boss in public." Because these beliefs are rarely tested, they persist even when leaders sincerely say they want input. Research on fear in organizations reaches a similar conclusion: fear of negative consequences is a major reason employees withhold information and ideas (Kish-Gephart et al., 2009).

Popular accounts attribute this to an "amygdala hijack." That story is a convenient metaphor but a weak foundation for a business case, and we do not rely on it here. The behavioral evidence is stronger and sufficient: people anticipate social costs and adjust what they say.

THE SCIENCE

What the evidence shows

51 teams

Edmondson, 1999

The foundational study

Edmondson studied 51 work teams at a manufacturing company and found that team psychological safety was associated with learning behavior, which in turn was associated with team performance.

136 samples

22,000+ individuals · nearly 5,000 groups · Frazier et al., 2017

The meta-analysis

Psychological safety was strongly related to engagement, job satisfaction and commitment, and moderately related to task performance. It was also strongly linked to learning behaviors such as information sharing. Most of the underlying studies were cross-sectional, so the findings show association rather than proof of cause.

180 teams

250+ attributes · Google, Rozovsky, 2015

Project Aristotle

Google's People Operations team reported that psychological safety was by far the most important of five dynamics that distinguished effective teams. Members of higher-safety teams were less likely to leave and were rated as effective twice as often by executives. This is a company report, not a peer-reviewed study, but it is widely cited and consistent with the academic literature.

Generative

Forsgren, Humble & Kim, 2018

Software delivery research

The *Accelerate* research program found that a "generative" culture, one in which information flows freely and failure leads to inquiry rather than blame, predicts better software delivery and organizational performance.

HONEST LIMITS

Three cautions keep the paper credible.

CORRELATION	Because much of the evidence is correlational, safety may be both a cause and a result of good performance.
NOT LINEAR	Recent work suggests the relationship between safety and performance may not be a simple straight line, which is a reason to pair safety with accountability (Eldor, Hodor & Cappelli, 2023).
NOT NICENESS	Edmondson's own framing is that high standards and high safety must coexist, because safety without accountability produces a comfortable but unambitious team.

Psychological safety is not niceness. The aim is candor, not comfort.

DIAGNOSIS

How silence shows up

Silence rarely appears as a single event. It appears as a pattern that costs time and money.

SYMPTOM IN OPERATIONS	WHAT YOU SEE	LIKELY BEHAVIORAL DRIVER
Late-surfacing bad news	Status stays green until a deadline is missed	Fear of being the bearer of bad news; belief that problems reflect on the messenger
Senior-voice dominance	Decisions follow the first or most senior opinion	Deference; implicit voice theories
Low idea volume	Few proposals from junior or new staff; retrospectives are polite	Fear of looking ignorant or disloyal
Repeat incidents	The same failure recurs under different names	Blame replaces learning; details are withheld
Hedged language	"It might be worth considering..." where a direct risk call is needed	Self-protection
Exit-interview revelations	"I saw this a year ago but did not think it would help to say so"	Futility and perceived retaliation risk

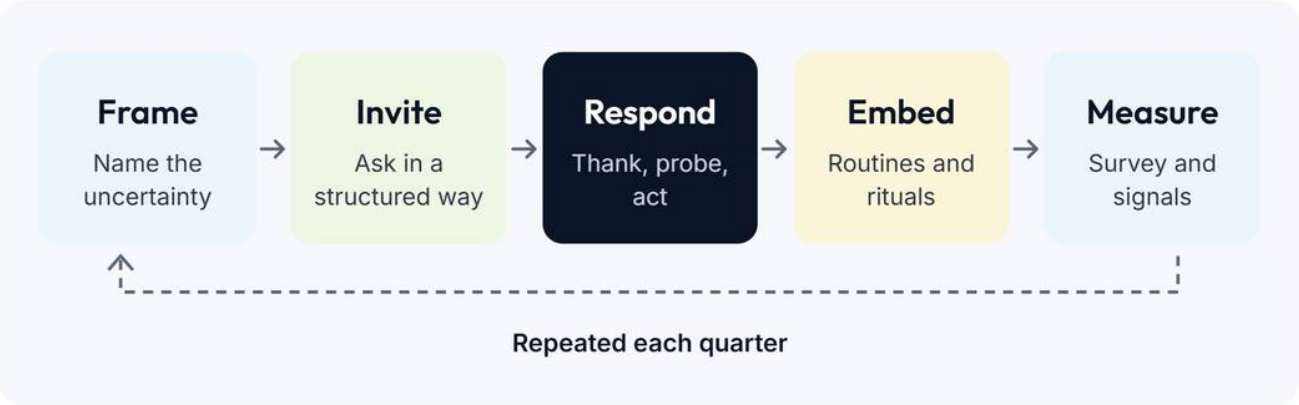
These symptoms are costly in innovation settings because they hide exactly the information that discovery requires: what is not working, what customers did not like and what assumptions are wrong.

THE METHOD

The Safety Engine

A repeating five-step loop that leaders run with their teams.

It treats safety as something built through behavior and structure rather than announced in a values statement.



The leader's response determines whether voice continues.

THE STEP THAT DECIDES EVERYTHING

The leader's first reaction to bad news teaches everyone whether speaking up is safe. A single punished messenger can undo months of effort.

THE SAFETY ENGINE

FRAME

Frame the work as learning

People speak up more when they understand that the work involves uncertainty and that their input is needed.

IN PRACTICE · Open each project with a short statement of what is known, what is not known and what could go wrong. Replace "we must execute flawlessly" with "this is a new problem, and we will only get it right by learning fast; I need to hear about problems early." Add a "what we do not know yet" section to project briefs. Edmondson's core point is that leaders set the frame by naming fallibility and stakes together.

INVITE

Invite input in a structured way

A general invitation ("any questions?") is easy to ignore. A specific, structured request makes speaking the default.

IN PRACTICE

- Ask specific questions: "What is the biggest risk you see in this plan?" rather than "Any concerns?"
- Collect written input before discussion, so the first and loudest voice does not anchor the room (a direct link to the nudge design in The Nudge Enterprise).
- Invite the most junior people to speak first and the most senior last.
- Rotate a designated challenger role so dissent becomes a task, not a personal risk.
- Provide an anonymous channel for topics people will not raise openly, and publish what was done with the input.

THE SAFETY ENGINE, CONTINUED

RESPOND

Respond productively

The leader's first reaction to bad news teaches everyone whether speaking up is safe.

IN PRACTICE · Train leaders in a three-part response:

Thank	Probe	Act or explain
for the specific information	"what led you to that conclusion? what would you do?"	what will change, or why nothing will

Close the loop visibly with a regular "you said, we did" update. Responses to bad news should be rehearsed, because the instinctive reaction under pressure is often defensiveness.

EMBED

Embed the practice in structure

Safety that depends on one charismatic leader disappears when the leader moves on. Routines and rituals carry it forward.

IN PRACTICE

- **Blameless post-mortems** that investigate how a system allowed an error, not who caused it (Allspaw, 2012). Keep a clear distinction between honest mistakes, which are learning opportunities, and reckless behavior, which still has consequences.
- **Intelligent-failure recognition:** celebrate well-designed experiments that did not work, and publish what was learned.
- **Pre-mortems** before major commitments (see The Sunk Cost Trap).
- **Stop-the-line norms** so anyone can pause work on a safety, quality or ethical concern.
- **Onboarding scripts** that tell new hires how and why to challenge ideas.

THE SAFETY ENGINE, CONTINUED

MEASURE

Measure and iterate

What is not measured drifts. Safety can be measured through short surveys and behavioral signals.

IN PRACTICE · Run a brief team survey each quarter using Edmondson's (1999) seven-item scale or a close adaptation, report results at team level only, and set a minimum group size for reporting so individual answers stay anonymous. Combine the survey with behavioral signals such as time to surface critical issues, number of issues and ideas raised and completion of post-mortems. Review results with the leader and the team, agree one improvement action and re-measure.

SAFETY IS NOT THE ABSENCE OF STANDARDS

Raise both safety and accountability

Psychological safety and accountability are separate dimensions. The goal is to raise both.

	LOW ACCOUNTABILITY	HIGH ACCOUNTABILITY
HIGH SAFETY	**Comfort zone** Pleasant, but little challenge or learning	**Learning zone** Candid, high-standard, fast-learning
LOW SAFETY	**Apathy zone** People disengage	**Anxiety zone** People comply, hide problems and avoid risk

CASE ILLUSTRATION

Halden Software

A composite illustration, not a report on a named organization. All figures are illustrative.

A hypothetical company with 160 people in 14 product and engineering teams. Two consecutive launches slipped by a quarter, and both post-mortems found issues that team members had noticed early.

56%
favorable on the safety items, with wide variation between teams

9 days
median for a critical issue to reach the leadership group

WHAT THE VP OF ENGINEERING AND CPO RAN OVER TWO QUARTERS

FRAME	Every project brief gained a "what we do not know" section, and leaders opened kickoffs by naming the uncertainty.
INVITE	Planning meetings began with written input, and a rotating challenger role was introduced.
RESPOND	Managers completed a short workshop on the thank, probe, act routine and practiced it with real examples. A monthly "you said, we did" note went to all teams.
EMBED	Blameless post-mortems became standard for incidents and missed milestones, and a quarterly "intelligent failures" review was added to the leadership meeting.
MEASURE	A quarterly survey and a dashboard of behavioral signals went to team leads. Safety scores were deliberately not used in individual manager evaluations, to avoid inflated answers.

HALDEN SOFTWARE, CONTINUED

TWO LESSONS

- ✓ A few managers initially treated the exercise as theatre; the teams noticed within weeks, and scores for those teams lagged until the managers changed how they responded.
- ✓ The anonymous channel produced little at first, until leaders published what they had done with the first submissions.

ILLUSTRATIVE RESULTS AFTER TWO QUARTERS

Favorable safety survey score **56% → 76%**	Median days for a critical issue to reach leadership **9 → 3**
Escaped defects per release (baseline = 100) **100 → 72**	Median release cycle **11 wks → 8 wks**
Annualized regretted attrition (trailing six months) **14% → 10%**	

The organization attributed the gains to earlier detection and cheaper fixes, not to any single workshop.

A caveat applies: with only two quarters of data and no control group, the team could not rule out that other changes contributed. That is why the measurement plan includes comparison groups.

MEASUREMENT

KPIs to track

Measure safety directly, and measure the outcomes it is supposed to improve. Also track a delivery or accountability measure, so that nobody can read the program as a lowering of standards. Baselines and targets are illustrative; replace them with your own.

KPI	DEFINITION	ILLUSTRATIVE BASELINE	TARGET (6 MONTHS)
Team safety score	Share of favorable answers on a seven-item survey	55–65%	75–85%
Time to surface critical issues	Median days from first awareness to leadership visibility	7–10 days	2–4 days
Voice rate	Issues and ideas raised per person per quarter	Measure at start	Up 30% or more
Post-mortem completion	Incidents with a blameless review within five working days	30–50%	90%+
Escaped defect rate	Defects found after release per release	Baseline = 100	70–80
Release cycle time	Median weeks from commitment to release	Measure at start	Down 15–25%
Regretted attrition	Voluntary departures of people the company wanted to keep	10–15% annualized	2–4 points lower
Delivery accountability Guardrail	Share of commitments met	Measure at start	Held or improved
Survey participation Guardrail	Response rate	Measure at start	Above 80%

THE BUSINESS CASE

A simple ROI model

The financial case can be built from three effects: rework avoided, retention improved and capacity regained through faster cycles. A transparent formula is more persuasive to a finance leader than a headline claim.

$$ROI = \frac{B_{\text{rework}} + B_{\text{retention}} + B_{\text{speed}} - C}{C}$$

B is the annual benefit from each effect and C is the annual program cost. Worked illustration for one 40-person engineering group; every input is an assumption to replace with your own data.

INPUT	ASSUMPTION	RESULT
Fully loaded cost per engineer	$120,000 per year	—
Capacity lost to rework from late-found problems	6% today; a one-third reduction (2 points)	0.8 FTE regained, worth $96,000
Regretted attrition	12% falls to 9%	1.2 fewer leavers
Replacement cost per leaver	0.75 times salary	$90,000 each, so $108,000
Speed benefit (earlier launches)	Not monetized in this example	Upside
Program cost	Training, facilitation, survey tooling, leader time	$70,000

Benefits	Costs	Illustrative ROI
$204,000	**$70,000**	**~191%**

Replacement-cost multiples in published sources vary widely, so test the sensitivity of the result with a lower and a higher figure. Treat the model as a planning tool, not a forecast. Because the benefits depend on correlational research, present them as expected rather than guaranteed, and validate with a pilot that includes a comparison group.

GUARDRAILS

Where the intervention goes wrong

Safety is not niceness. Teams that confuse safety with agreeableness stop challenging each other. Keep standards high, set clear expectations and give direct feedback; the aim is candor, not comfort.

Survey without action. Asking people about safety and then doing nothing proves that speaking up is futile. Commit to sharing results and acting on at least one finding before the first survey.

Performative vulnerability. Leaders who announce that they welcome challenge and then react badly to it do more harm than leaders who made no announcement. Behavior in the first moment of bad news is the real signal.

Gaming the scores. If safety scores feed into manager pay or promotion, answers inflate and trust erodes. Use scores for learning at team level, not for individual ranking.

Local safety, global silence. A team can be safe internally while people stay silent toward other functions or senior leaders. Check cross-team and upward voice, not only within-team scores.

Overclaiming causation. Most of the evidence is correlational. State the limits plainly in any business case, and run a pilot with a comparison group before scaling.

LEGAL AND COMPLIANCE NOTE

Anonymous channels and surveys carry obligations. Promise confidentiality only where you can deliver it, and be clear about what happens to reports of misconduct. In the European Union, the Whistleblower Directive (2019/1937) and the national laws that implement it set requirements for reporting channels and protection against retaliation, and these apply regardless of the team-level safety program. Survey data about employees falls under data protection law, so use aggregated reporting, a minimum group size and a documented lawful basis. Involve legal and HR before launching, and keep clear the boundary between psychological safety practices and formal grievance or investigation procedures. This paper is not legal advice.

THE PLAN

The first 90 days

DAYS 1–30

Diagnose and prepare

- Choose two or three pilot teams and a comparable group of teams to serve as a comparison.
- Run a baseline survey (seven items, team-level reporting, minimum group size) and gather behavioral baselines such as time to surface issues.
- Interview five to eight people across levels about what is hard to say and why.
- Brief the sponsoring executive on the Respond step; their reaction to the first piece of bad news sets the tone.
- Complete the legal and data-protection review of the survey and any anonymous channel.

DAYS 31–60

Launch the loop

- Add "what we do not know" to project briefs and begin written-first input in planning meetings.
- Train pilot-team leaders in thank, probe, act and rehearse with real scenarios.
- Run the first blameless post-mortem and publish the lessons.
- Open the anonymous channel and publish the first "you said, we did" update.

DAYS 61–90

Embed and measure

- Re-run the survey and compare pilot teams with the comparison group.
- Review behavioral signals and update the ROI model with real numbers.
- Add the successful routines to meeting templates, onboarding and the leadership agenda.
- Decide on scale-up, and set a quarterly review so safety stays on the agenda.

CALL TO ACTION

Pick one team that is working on a risky, uncertain project and run the first 30 days. You will learn more from watching how the leader reacts to the first piece of bad news than from any survey. Start there, protect the person who delivers it and make sure everyone sees what happens next.

CONTINUE READING

Connections to the series

The Sunk Cost Trap	Pre-mortems and honest project kills only work when people feel safe saying a project should stop.
The Nudge Enterprise	Structured turn-taking and written-first input are choice-architecture tools applied to meetings.
Atomic Habits at Scale	The leader's response to bad news must become a habit, not a good intention.
The Halo and the Horn	Reviews that punish the bearer of bad news teach everyone to stay quiet.
Proximity Bias and the Hybrid Divide	Remote and hybrid team members often have the least safe routes for voice.
Beyond the Bonus	Autonomy and mastery flourish where people can admit what they do not yet know.

Brain Interacts

ABOUT BRAIN INTERACTS

Brain Interacts is a team analytics and behavioral analysis company. We build Corporate Brain and work with organisations on culture studies, root-cause analysis, and behavior-led operating design.

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SOURCES

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